Tree Height Triangulation Shootout

Measure a tree without climbing it.

Win condition: Closest to instructor reference height.

THE SCIENCE YOU ARE USING

Angles give height. You can find a tree's height without touching the top. Sight the treetop through a paper clinometer to read an angle, measure your distance to the trunk, and the geometry of a right triangle gives the height. This is remote sensing with a protractor.

Accuracy from method. Small sighting errors grow over distance, so careful technique matters: level your eye, sight steadily, measure distance honestly. The team closest to the instructor's reference height wins.

HOW TO BUILD AND RUN IT

01 **Build the clinometer.** Fold a paper clinometer with a weighted string plumb line to read angles.

02 **Pace a known distance.** Measure your straight-line distance from the trunk and record it.

03 **Sight the treetop.** Look along the clinometer at the very top and read the angle steadily.

04 **Compute the height.** Use the angle, your distance, and your eye height to estimate the tree height.

05 **Check against reference.** Compare to the instructor reference. The closest estimate wins.

Safety: Outdoor boundaries. Allergy: Pollen sensitivity.

MATERIALS

Paper clinometers	per team
String and washers	shared
Rulers and tape measures	shared
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

The ONE variable we are testing:

Baseline result: ____________________

After redesign: ____________________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

If you ran it again, what is the ONE thing you would change? ____________________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	**100**